

Vital Sleep Patch:

Transforming Pediatric Sleep Apnea Diagnosis through Home-Based Testing

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Week 5 Interviews Conducted: 11 human
Total Interviews: 49 human + 10 synthetic

★ **Problem Solving Journey:** Started with a wearable sensor patch and AI-based diagnostic pipeline for pediatric OSA. Now, through customer discovery, we are focusing on our customer relationship strategy ★



Distribution channel order of operations



**Partner with
AMCs**



**Partner with online
test providers**



**Direct to clinicians
and consumers**

Model

Sell to AMCs as a triage
test to support PSGs

Sell to online providers
(they sell to consumers
with long-term care)

Sell directly to
independent clinicians and
consumers

For
consumers

Get from waitlist; **keep**
for tests; **grow** via
downstream therapies

Get from referrals; **keep**
for tests; **grow** via repeat
tests and long-term care

Get from outreach; **keep**
for tests; **grow** via repeat
tests and long-term care

Hurdles

Reimbursement / payor
pathway; data <>
workflow integration

Reimbursement / payor
pathway; workflow
integration

Marketing (25% of
revenue) and category
creation for pediatric OSA

Estimated economics [AMCs]

700k

LTV

10x

LTV : CAC

~12 months

CAC Payback

~85%

Gross Margin

\$2,000 / device

Price / Unit

\$300 / device

COGS

**\$70,000 ACV (35
devices/AMC -> 1k
tests)**

Avg Revenue

\$70,000

CAC

10% annual

Annual Churn

10 years

Avg Customer Life

12–18 months

Sales Cycle

✓ Strengths

- ▶ No existing vendor, can refine pricing model / contract structure
- ▶ Sticky customer with low churn
- ▶ Aligns w/ our core capability (clinical grade accuracy)

✗ Watch-outs

- ▶ Long sales cycle (12–18 mo)
- ▶ High CAC
- ▶ VAC approval bottleneck

Est. economics [Online test providers]

\$1M

LTV

27x

LTV : CAC

~2 months

CAC Payback

~87%

Gross Margin

\$150 / test (wholesale)

Price / Unit

~\$20 / test (reusable device)

COGS

\$360k ACV (200 tests/mo/partner)

Avg Revenue

\$40,000

CAC

~30% annual

Annual Churn

3 years

Avg Customer Life

3–6 months

Sales Cycle

✓ Strengths

- ▶ Massive LTV per partner
- ▶ Near-zero churn
- ▶ Low CAC vs. revenue

✗ Watch-outs

- ▶ Few partners = concentration risk
- ▶ Margin dependent on reuse model
- ▶ Partner must have pediatric reach

Est. economics [D2C]

**\$36k (clinic) ·
\$800 (D2C)**

LTV

**12× (clinic) ·
5x (D2C)**

LTV : CAC

1-3 months

CAC Payback

**85% (clinic) ·
25% (D2C)**

Gross Margin

**\$2,000 / device (clinic)
· \$400 OOP (D2C)**

Price / Unit

\$300 / device

COGS

**\$12,000 ACV/clinic (6
devices / clinic) ·
\$400 one-time (D2C)**

Avg Revenue

**\$3,000 (clinic) ·
\$175 (D2C)**

CAC

**~30% annual (clinic) ·
100% (D2C)**

Annual Churn

**~3 yrs (clinic) ·
2 tests in 1y (D2C)**

Avg Customer Life

1–3 months

Sales Cycle

✓ Strengths

- ▶ Low CAC vs. LTV for clinics
- ▶ Fast sales cycle
- ▶ Validates beachhead demand

✗ Watch-outs

- ▶ High SMB churn
- ▶ Requires high-volume acquisition
- ▶ D2C economics negative



D2C economics are currently negative — not recommended standalone

Channel Economics — Summary

All figures are estimates based on industry benchmarks and current pricing assumptions

Metric	AMCs (Bulk Direct)	Online Providers (OEM/B2B2C)	Clinicians & D2C (Direct)
Price / Unit	\$2,000 / device	\$150 / test (wholesale)	\$2,000/device · \$400 D2C
Gross Margin	85%	87%	85% (clinic) · 25% (D2C)
CAC	\$70,000	\$40,000	\$3,000 (clinic) · \$175 (D2C)
Customer LTV	700k	\$1M	\$36k (clinic) · \$800 (D2C)
LTV : CAC	10x	27x	12x (clinic) · 5x D2C
CAC Payback	~12 months	~2 months	1-3 months
Annual Churn	~10%	~30%	~30% (clinic) · 100% (D2C)
Sales Cycle	12–18 months	3–6 months	1–3 months

Strategic note: AMC channel aligns with clinical accuracy positioning despite lower LTV:CAC. Online providers offer best pure economics but require a reusable device model. Clinic direct shows strong unit economics; D2C alone is unprofitable.

Focus on Academic Medical Centers

We need a foundation to scale over immediate profitability.

Source of Truth

Validations by AMC clinicians fuels further trust in our technology & platform.

Payor Pipeline

Efficacy in AMCs will underpin approval & recommendation by Academies and eventual coverage by payors.

Scaling Opportunity

Academic medical centers don't currently use pediatric home tests. Propagating through AMCs nationwide would provide a tremendous clinician customer base and moat.

These aren't true in the reverse direction (tech enabled service providers → AMCs).

Get / Keep / Grow Hypotheses (AMC Focus)

GET

How we acquire AMC customers

Backlog relief acquisition:

We can get AMC sleep clinics onboard by positioning the device as a way to reduce pediatric PSG backlog and shorten time to diagnosis



Patient volume acquisition:

We can acquire AMC sleep clinics by helping serve more pediatric patients (increase patient volume) through at-home testing, increasing diagnostic throughput beyond the limits of in-lab PSG capacity (remove bottlenecks)



KEEP

How we retain AMC customers

Embed into the sleep clinic's operating workflow:

Make at-home testing part of routine intake by integrating ordering, parent setup, data upload, scoring, and reporting into the clinic's existing data review pipeline



Build trust through high quality clinical validation:

Retain clinics by proving pediatric accuracy against PSG and addressing key clinician concerns like sensor reliability and movement artifacts



GROW

How we expand lifetime value

Add repeat-use monitoring:

Provide post-treatment follow-up value for patient monitoring without requiring repeated in-lab sleep studies



Expand within the same AMC (other specialties):

Move from the sleep clinic to deployment directly into ENT, pulmonology, obesity, neuromuscular, and other pediatric referral clinics



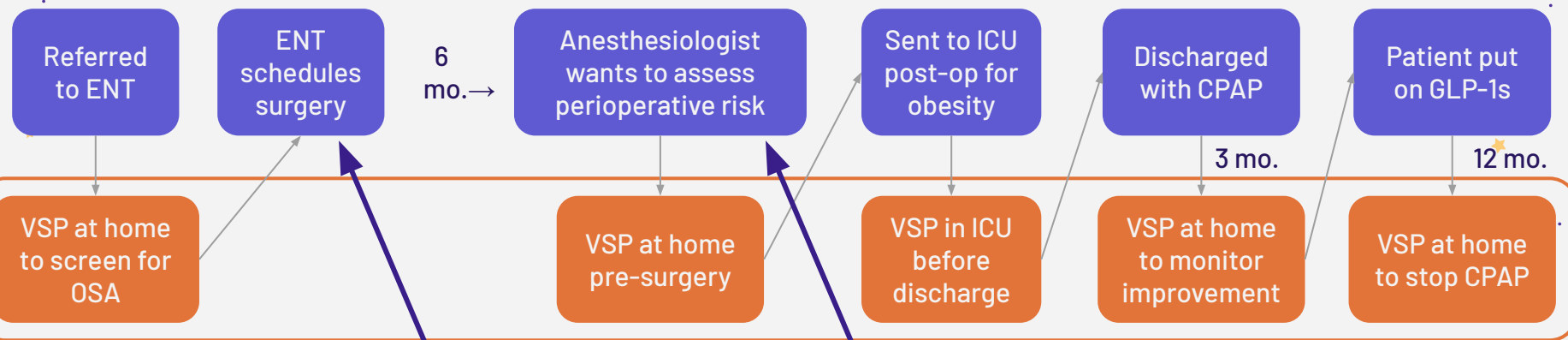
Financial Incentive: Increased number of patients → increased revenue for sleep clinics (and us)

Our unique value prop: Quality focused → can be deployed in pediatric cohort

The New Patient Journey (via AMC) ★

Maya's PCP has referred her to an ENT for suspected sleep apnea.

But now, Stanford Medicine uses Vital Sleep Patches (VSP) across their practices.



The Triple Win: Get patients earlier. Keep patients. Grow value capture (for us and AMC).

all by facilitating gold-standard care delivery.

Invalidated Hypothesis: Canadian AMCs

Context:

- March 2026: **Canadian Thoracic Society & Canadian Sleep Society** issued a nationwide recommendation: *"Level 3/Home Sleep Apnea Testing is recommended as an alternative option to diagnose OSA in otherwise healthy children over 5-years of age for whom access to polysomnography is effectively absent...Given patient and family preferences for testing at home, further investment is needed in developing accurate and accessible home-based testing options to diagnose OSA in children."*

Hypothesis:

- We could focus on Canadian Academic Medical Centers as initial rollout partners for our clinical studies and patient deployment

Data Collected from Interviews:

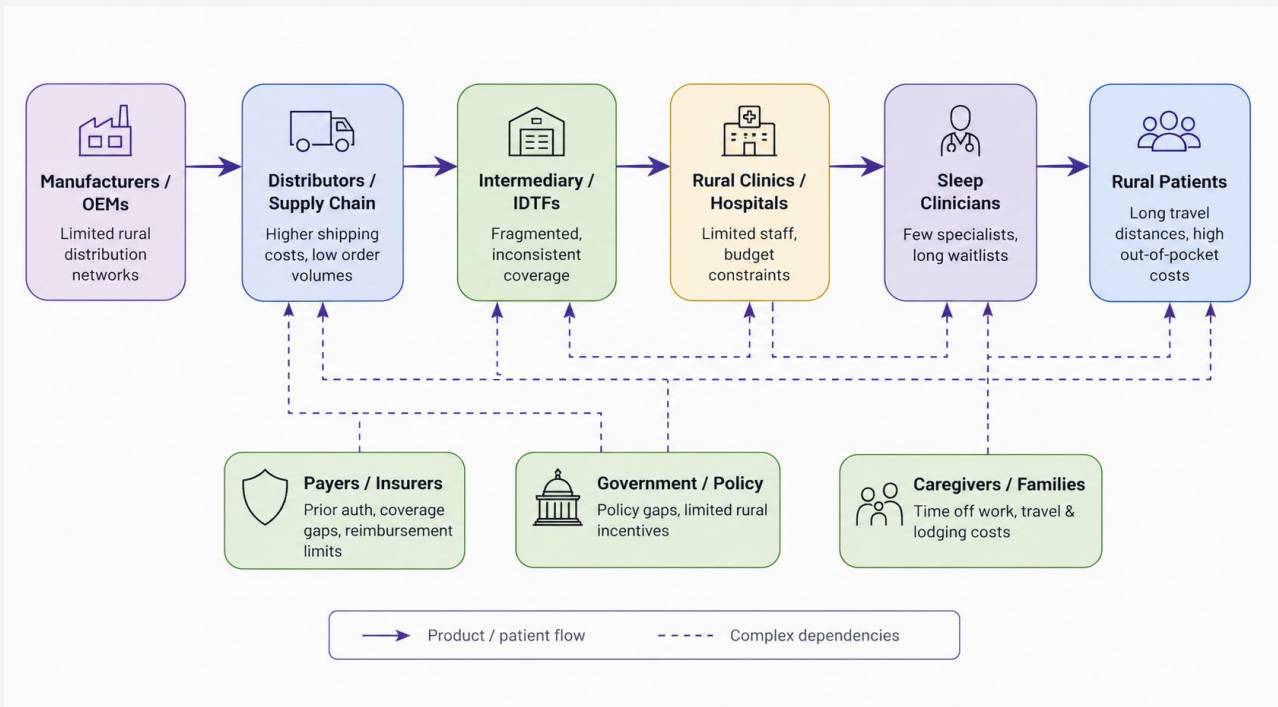
- **D2C model would have low demand;** strong Canadian patient preference for default zero-cost options
- **Each province/territory has separate approvals process** that would need FDA certifications and economic analyses for cost-effectiveness
- Incentive structures better aligned than in US, with doctors incentivized to get through backlogs

Decision:

- Would be a good eventual market, but can't sustain a business with Canada as our beachhead

Invalidated Hypothesis: Rural Distribution

Many players, touchpoints, and barriers make this path difficult and slow. We will be **deprioritizing** this channel for the moment.



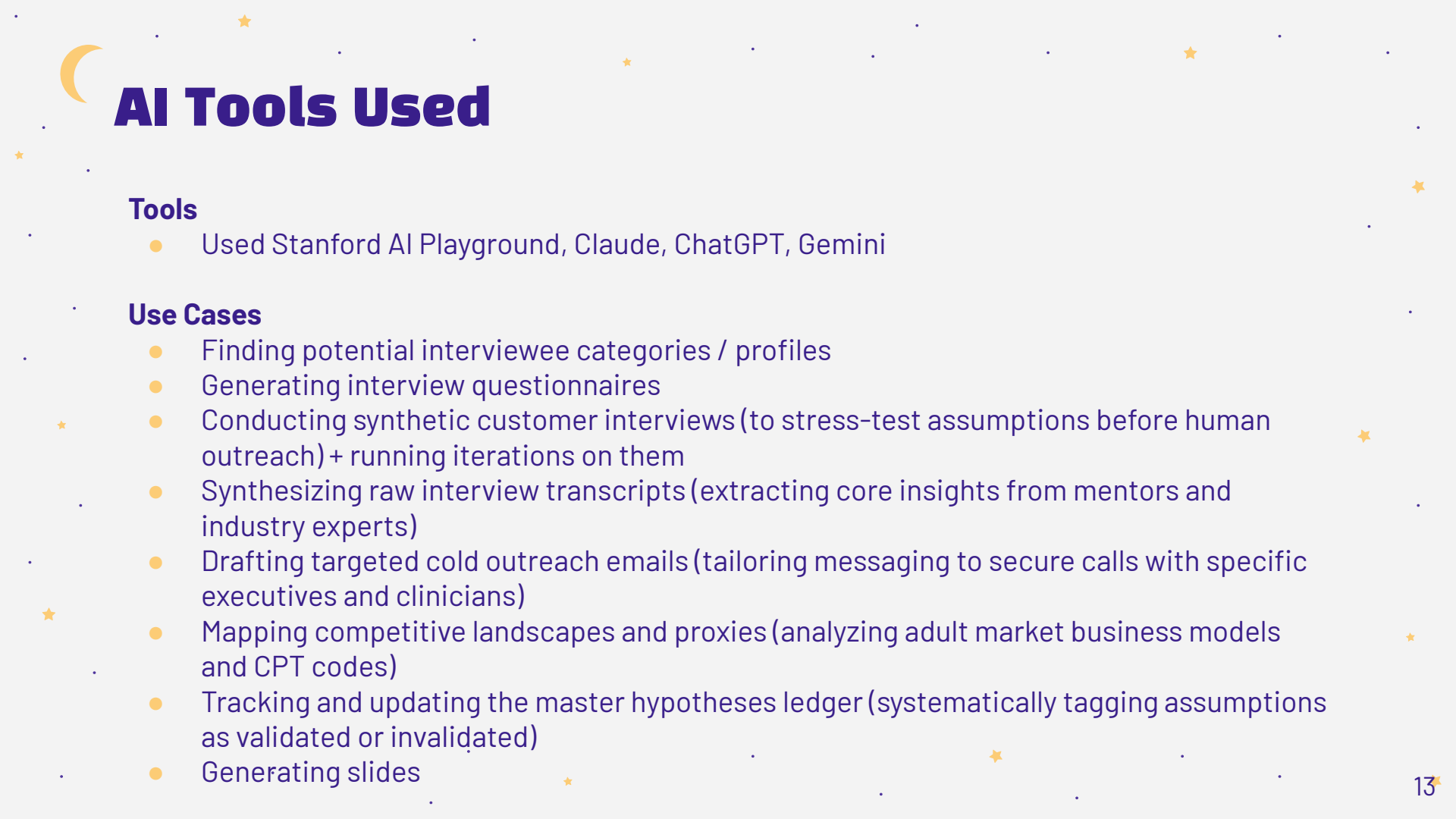


To-do Next Week and Help needed!

- Continue validating **regulatory and reimbursement pathway** (in touch with Treehub, Biodesign)
- Focus on deep diving into:
 - Academic Medical Centers: Validate value proposition and test economic assumptions; co-validation against PSG, clear clinical guidelines for when to use Vital vs. full sleep study, integration with referral workflows
 - Through online test providers: Keep pressure testing economics
- Build a **financial value prop model**

Help needed

- Intros to other AMCs
- Intros to regulatory experts and payors



AI Tools Used

Tools

- Used Stanford AI Playground, Claude, ChatGPT, Gemini

Use Cases

- Finding potential interviewee categories / profiles
- Generating interview questionnaires
- Conducting synthetic customer interviews (to stress-test assumptions before human outreach) + running iterations on them
- Synthesizing raw interview transcripts (extracting core insights from mentors and industry experts)
- Drafting targeted cold outreach emails (tailoring messaging to secure calls with specific executives and clinicians)
- Mapping competitive landscapes and proxies (analyzing adult market business models and CPT codes)
- Tracking and updating the master hypotheses ledger (systematically tagging assumptions as validated or invalidated)
- Generating slides

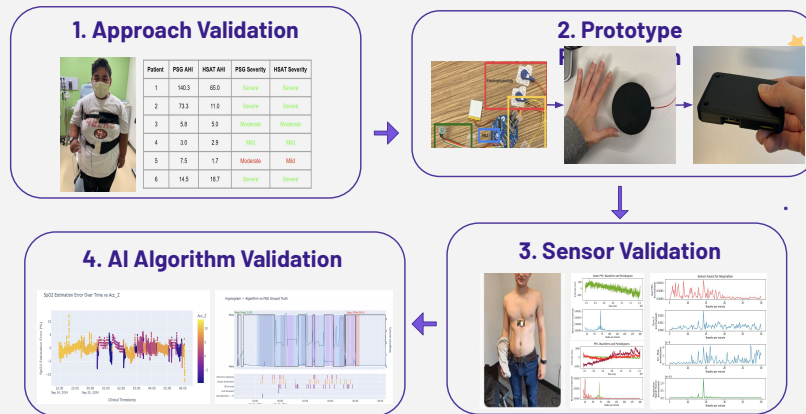
Experiments

Device MVP:

- Testing whether our current prototype delivers the exact sensor data and workflow simplicity required to trigger adoption.
- Spec comparison between our device and competitors

Financial Model and Digital MVP:

- Financial validation model for AMCs (how much money will they be making?)
- Refining market size model for the product
- Landing page for consumers to see initial interest (“waitlist”)



Vital Health — Pediatric OSA Market Sizing (TAM / SAM)			
Metric	Formula Logic	2026	2027
Pediatric AMC Tests with no supply constraints	# Pediatric tests per year / Diagnosis rate		
Accessible undiagnosed patients (channel filter)	Undiagnosed × Assumptions1SC\$22 (channel reach %)	876,000	
Annual new-diagnosis rate (tests/yr)	Accessible patients × annual dx rate	438,000	
SAM — Annual revenue potential (\$M)	Annual tests × device price / \$1M	\$3,379.2	
SOM — Serviceable Obtainable Market (Vital Health Realistic Capture)			
Vital Health market share assumption	See Assumptions tab — SOM Peds rows	0.3%	
Tests captured by Vital Health (units)	SAM tests × market share	1,314	
SOM — Revenue (\$M)	Captured tests × device price / \$1M	\$0.1	
SOM — Gross Profit (\$M)	Captured tests × (price – COGS) / \$1M	\$2.5	
SOM / SAM ratio (penetration check)	SOM revenue + SAM revenue	0.0%	